

Srevarshan Subramanian

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PROFESSIONAL SUMMARY

AI/ML Engineer experienced in fine-tuning open-weight LLMs and VLMs (LoRA) to solve real-world document intelligence and multimodal challenges. Accomplished Springer researcher and holder of Indian patent, with research-oriented mindset to reproduce research papers, conduct thorough experiments, and produce models instead of relying on APIs.

EDUCATION

B.Tech – Computer Science Engineering (AI & ML)
SRM Institute of Science and Technology, Trichy

2023 – 2027

CGPA: 9.73 / 10

Available for a 10–12 month full-time internship (2027 Graduating Student)

TECHNICAL SKILLS

Deep Learning: PyTorch, transformer architectures, model training, Fine Tuning LLM, Lora

Gen AI: Vision and Small Language Models, RAG pipelines, LangChain, Vector DB, Ollama

ML Frameworks: Hugging Face, Scikit-learn, ML Algorithms

Backend & Deployment: FastAPI, Docker, Amazon web services, Git and GitHub

Edge AI / Embedded: Raspberry Pi, NVIDIA Jetson Orin Nano

Programming Languages: Python, SQL

EXPERIENCE

AI Product Developer | EmedLogix

Nov 2025 – Jun 2026

- Developed a **production-grade AI-powered healthcare credentialing platform** for a US healthcare client, hosting the full-stack application on AWS.
- Engineered automated **OCR-based document processing and Auto filling Fields**, integrating REST APIs, performing data validation and transformation, and storing structured doctors records in PostgreSQL, reducing manual work by **90%**.
- Deployed backend services on **AWS Elastic Beanstalk** and provisioned the frontend with **AWS Amplify**, ensuring scalable, reliable and Secure production deployments

AI/ML Intern | Microsoft × Edunet Foundation

May 2025 – Jun 2025

- Developed **AI Nose**, an **ML-powered real-time atmospheric odor classification system** using **ESP32, MQ3/MQ9 gas sensors**, and a **Random Forest classifier**.
- Engineered an entire machine learning pipeline by continuously acquiring gas sensor readings, preprocessing and labeling data into **four odor classes (Clean, Mild, Foul, and Very Foul)** for supervised model training.
- Trained and optimized a **Random Forest** model achieving **98% classification accuracy**, and deployed it in a **real-time inference pipeline** that continuously fetches real time data from RTDB and classifies live sensor readings from the ESP32.
- Integrated the prediction script with **Firestore** for real-time monitoring, visualization, and automated sanitation alerts based on predicted odor levels.

PROJECTS

1) Nutri-Mind: Tailored Nutrition Analysis with Vision and Large Language Model | (Research Paper Accepted)

- Conducted months of research on **Generative AI and LLMs** before discovering the idea of developing a product to inform users about the safety of pre-packaged products based on their medical history.
- Built an **OCR pipeline using VLM** that retrieves nutrition-label information from product images by evaluating the performance of **Pytesseract**, **PaddleOCR**, and **EasyOCR** and came to conclusion using **Qwen2.5-VL**, which performed the best in real-world scenarios
- Used the extracted nutrition information and user's medical history to produce structured JSON data and devised a prompt template that merged both data pieces.
- Created a **JSON dataset** with more than 5,000 rows of data by obtaining and mapping relevant public datasets from **FSSAI** and the **US food database** and **fine-tuned Mistral-7B** model with **LoRA** and **Unsloth** using the **Google Colab T4 GPU**.
- The tuned model gives a **food-safety recommendation** to the user according to their condition with a **safe consumption level and risk** of overuse.
- The submission "[Nutri-Mind: Personalized Nutritional Analysis Using Vision and Small Language Models](#)," was accepted in the **Springer Conference Proceedings (ICDAM 2026)** by the **London Metropolitan University** and ranked in the **top 20%** out of over 2000 submissions worldwide.

2) Banana Weevil Early Detection System (Patent Published Research Project | Edge AI Jetson Nano)

- Created **dual-modal acoustic sensing device using bio-acoustic detection and percussive** resonance techniques giving the capacity of spotting inside weevil infection of bananas.
- **MFCC feature extraction** from RAW Audio and pass to **MobileNetV2 CNN** that gives output and classification of 5 stages of the infestation using data with almost 80% accuracy, working entirely offline on the Jetson Orin Nano.
- Set up CUDA and cuDNN on NVIDIA Jetson Orin Nano to enable GPU-powered inference for the audio capture, cleaning, feature extraction, and CNN inference pipeline.
- This Invention significantly decreased banana crop loss with the help of the early detection of the infestation, created low latency and cost and cloud-free device for remote field usage.
- **Patent granted:** "[Device for Banana Weevil Detection through Dual-Mode Acoustic Sensing](#)".

RESEARCH AND PUBLICATIONS

- **Research Paper (Accepted)** – "[Nutri-Mind: Personalized Nutritional Analysis Using Vision and Small Language Models](#)" – **Springer Proceedings, 7th International Conference on Data Analytics and Management (ICDAM 2026)**, organized by **London Metropolitan University** (*Top 20% of 2,000+ global submissions*).
- **Book Chapter (Published)** – "[Sevai Kural: A Smart Voice-Based System for Better Public Services](#)", published in the **Proceedings of the Modern Tamil Computing Conference 2025**. (on the theme "*Advancing Tamil through Language Technology: From Linguistic Foundations to Large Language Models (LLMs)*".")

PROFESSIONAL CERTIFICATIONS

- Fundamentals of Deep Learning – NVIDIA DLI
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI / Stanford (2024)

ACHIEVEMENTS & RECOGNITION

- 1st Place – **Spectrum25 National level Hackathon** (VIT Chennai)